

# AI-Based Early Disease Risk Prediction System

## User Manual

Welcome to the **AI-Based Early Disease Risk Prediction System**. This application support Machine Learning techniques to predict the risk of **early Diabetes** and **Cardiovascular Disease** using user-provided health parameters.

The system is designed to assist users by providing an early risk assessment along with health-related recommendations. This manual provides a comprehensive guide for installing the required software, configuring the project environment, and running the application successfully.

## 1. System Requirements

Before using the application, ensure that the following software is installed on your computer.

### 1.1 Python Installation

The project requires **Python 3.9 or later**.

#### Step 1: Download Python

1. Visit the official Python website: <https://www.python.org/downloads>
2. Download the latest stable version compatible with your operating system.

#### Step 2: Install Python

For Windows users:

- Run the downloaded installer.

- **Important:** Check the option “Add Python to PATH” before clicking **Install Now**.

### Step 3: Verify Installation

Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:

```
python --version
```

Expected output: Python 3.10.x

or any version greater than 3.9.

## 1.2 Installing Visual Studio Code (VS Code)

Visual Studio Code is the recommended Integrated Development Environment (IDE) for this project.

### Download VS Code

1. Visit: <https://code.visualstudio.com>
2. Download the installer for your operating system.
3. Complete the installation process by following the on-screen instructions.

### Recommended Extensions

Install the following VS Code extensions:

- Python (Microsoft)
- Jupyter (Microsoft)

These extensions provide support for Python development and notebook execution.

## 2. Project Setup

### 2.1 Obtain the Project Files

Ensure that the project folder named: **FYRP** is available on your local machine.

You may place the folder in any desired location such as:

- Desktop
- Documents
- Development Workspace

### 2.2 Open the Project in VS Code

1. Launch Visual Studio Code.
2. Select: File → Open Folder
3. Browse to the project directory.
4. Select the **FYRP** folder and open it.

### 2.3 Create a Virtual Environment (Recommended)

Using a virtual environment prevents dependency conflicts with other Python projects.

#### Create the Virtual Environment

Open a new terminal in VS Code and execute:

```
python -m venv .venv
```

#### Activate the Environment

##### *Windows*

```
.venv\Scripts\activate
```

##### *macOS/Linux*

```
source .venv/bin/activate
```

When activated successfully, the terminal prompt will display: `(.venv)` indicating that the virtual environment is active.

### 3. Installing Project Dependencies

The application depends on several third-party Python libraries for machine learning, data processing, visualization, and web application development.

With the virtual environment activated, execute:

```
pip install streamlit pandas scikit-learn xgboost optuna imbalanced-learn
seaborn matplotlib joblib tqdm
```

#### Dependency Overview

| Library          | Purpose                                   |
|------------------|-------------------------------------------|
| Streamlit        | Builds the interactive web interface      |
| Pandas           | Data manipulation and analysis            |
| Scikit-Learn     | Core machine learning framework           |
| XGBoost          | High-performance gradient boosting models |
| Optuna           | Hyperparameter optimization               |
| Imbalanced-Learn | Handles class imbalance using SMOTE       |
| Seaborn          | Statistical data visualization            |
| Matplotlib       | Plotting and graph generation             |
| Joblib           | Saving and loading trained models         |
| TQDM             | Progress bars for long-running tasks      |

**Note:** Installation time may vary depending on internet speed and system specifications.

## 4. Running the Application

The user interface is developed using **Streamlit**.

### Step 1: Open Terminal

Ensure:

- The terminal is located at the project root directory.
- The virtual environment is activated.

### Step 2: Start the Application

Execute:

```
streamlit run app/app.py
```

### Step 3: Access the Application

After successful startup, Streamlit will automatically open a browser window.

Default local URL:

```
http://localhost:8501
```

### Application Navigation

The sidebar provides access to the following sections:

Home: Provides an overview of the project and system functionality.

Diabetes Prediction: Allows users to enter health metrics and obtain diabetes risk predictions.

Cardiovascular Prediction: Allows users to enter health metrics and obtain cardiovascular disease risk predictions.

Model Information: Displays information regarding the machine learning models used within the system.

## Generating Predictions

1. Navigate to the desired prediction module.
2. Enter the required medical parameters.
3. Click: **Predict Risk**
4. View:
  - Predicted disease risk
  - Confidence information
  - Personalized health recommendations

## 5. Machine Learning Model Development (Optional)

This section is intended for developers, researchers, or students who wish to understand the model training process.

### Requirements

Ensure the Jupyter extension is installed in VS Code.

### Available Notebooks

Navigate to:

notebooks/

and open:

## Diabetes Model

`diabetes_model.ipynb`

## Cardiovascular Model

`cardiovascular_model.ipynb`

These notebooks demonstrate:

- Data preprocessing
- Feature engineering
- Data balancing using SMOTE
- Hyperparameter optimization
- Model training
- Model evaluation
- Exporting trained models

Generated models are saved into the:

`models/`

directory.

## 6. Project Structure

The project directory is organized as follows:

text

```
FYRP/
├── .venv/                # Virtual environment folder (created by you)
├── app/
│   └── app.py            # The main Streamlit web application file
├── data/
│   ├── cardiovascular.csv # Dataset for Cardiovascular disease
│   └── diabetes.csv       # Dataset for Diabetes
├── models/
│   ├── cardiovascular_model.pkl # Trained ML model artifact
│   └── diabetes_model.pkl       # Trained ML model artifact
└── notebooks/
    ├── cardiovascular_model.ipynb # Training script/notebook
    └── diabetes_model.ipynb       # Training script/notebook
```

## Directory Description

app/ Contains the main Streamlit application.

data/ Stores datasets used during training and experimentation.

models/ Stores trained machine learning model files.

notebooks/ Contains notebooks used for model development, training, and evaluation.

## 7. Troubleshooting Guide

Issue: “python is not recognized”

Solution

- Ensure Python was installed correctly.
- Verify that Python was added to the system PATH.
- Try:

```
python3 --version
```

instead of:

```
python -version
```



## **Issue: ModuleNotFoundError**

### **Solution**

Ensure:

1. Virtual environment is activated.
2. Required libraries have been installed successfully.

**Run:** `pip install -r requirements.txt`

or reinstall the missing package.

## **Issue: Port 8501 Already in Use**

### **Solution**

If port 8501 is occupied, Streamlit will automatically select the next available port such as:

`http://localhost:8502`

Use the URL displayed in the terminal output.

## **Conclusion**

The AI-Based Early Disease Risk Prediction System provides an accessible and efficient platform for early disease risk assessment using machine learning techniques. By following the instructions outlined in this manual, users can successfully install, configure, and operate the application for both educational and practical healthcare prediction purposes.

For further customization or development, users are encouraged to explore the included Jupyter notebooks and machine learning workflows.